

Data Structures — End Semester Examination (Sample) 2024

Part A — Short Answer Questions (2 marks each)

1. Differentiate between an array and a linked list.
2. Define a stack and give one real-world application.
3. What is a binary search tree?
4. State the worst-case time complexity of quicksort.

Part B — Long Answer Questions (10 marks each)

5. Explain the working of merge sort with an example, and derive its time complexity.
6. Describe the operations of insertion and deletion in a binary search tree.
7. Compare stacks and queues in terms of structure and typical use cases.